Will Palaia

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Education

University of Massachusetts Amherst — Amherst, MA
Bachelor of Science in Computer Science, Bachelor of Arts in Economics — *February 2026*

- Coursework: Machine Learning, Algorithms, Database Management, Data Structures, Computer Systems

Skills

Languages: Python, Java, C, SQL (PostgreSQL, AWS Athena), TypeScript, JavaScript, HTML/CSS, VBA, R
Technologies: React, Node.js, FastMCP, Starlette, Docker, Kubernetes, AWS (Athena, S3, ECS), Git, Linux, Jira

Experience

Wellington Management — July 2026 – Present
Software Engineer — *Boston, MA*

- Engineered a config-driven Model Context Protocol (MCP) server in Python, FastMCP, and Starlette over 20+ AWS Athena data lake tables ($1.3T+ AUM), using on-demand schema discovery and sqlglot AST validation to cut LLM token overhead by 85% and reduce ad-hoc data lookup time by 90%
- Built an AI recommendation engine in Python (FastMCP) on AWS Athena, Salesforce, eVestment and Morningstar data that matches client needs to 252 investment strategies ($915B AUM) with cited evidence, cutting research time from hours to under 30 seconds

Software Engineer Intern — June 2025 – August 2025

- Built an interactive Python dashboard with Pandas and IPyWidgets that processes streaming bond data throughout trading hours, providing 70 fixed income team members with consolidated market analytics and trend visualizations
- Created a chatbot adaptor on internal AI platform that enables auto-generation of custom bots in minutes using only a keyword; deployed a Salesforce bot trained on 13,000+ internal documents, reducing manual build time by 95%

Imago Rehab — September 2024 – January 2025
Software Engineer Intern — *Lowell, MA*

- Cut customer support response time by 40% and handled 80% of common patient queries by integrating an AI-powered chatbot using Google Dialogflow

ISO New England — June 2024 – August 2024
Software Engineer Intern — *Holyoke, MA*

- Implemented an internal tool to automate transformer energy level verification and dynamic grid modeling in the New England energy grid, improving daily forecast accuracy by 6% and cutting manual analysis by 40+ hours/week
- Built a custom validation engine in pure Python that flagged data discrepancies with 100% accuracy across 37,000-row Excel datasets

BigBear.AI — June 2022 – August 2022
Software Engineer Intern — *Lexington, MA*

- Delivered a production-ready Jira web app to the Atlassian Marketplace using React, OAuth2, and agile workflows, improving project tracking visibility for distributed teams
- Architected and deployed Python APIs in Docker containers via AWS ECS, enabling high availability and scalability
- Boosted test coverage by 35% through custom-built unit tests and a full CI/CD pipeline using GitHub Actions

Projects

Prediction Market Arbitrage | *Python, aiohttp, REST APIs* — April 2026

- Engineered a Python-based trading engine utilizing aiohttp for asynchronous REST API polling to identify 1%–3% price discrepancies between Polymarket and Kalshi; implemented concurrent execution logic that achieved sub-500ms latency from signal to order, generating a 7% monthly return on capital with an 86% trade success rate

Extracurriculars & Interests

Minutemen International Equity Fund @ UMass Amherst — February 2023 – February 2026
Quantitative Software Engineer — *Amherst, MA*

Interests: Algorithmic trading, art history, poker, weightlifting, tennis, golf, drumming, video games